

MURMANSK, Russian Federation

WMO: 221130

Lat: **68.9558N**Lon: **33.0428E**Elev: **81**StdP: **100.35**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.0	-29.6	-36.8	0.1	-32.8	-32.9	0.2	-29.3	12.7	-6.9	11.0	-5.9	1.4	180	0.766

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	24.8	16.3	22.1	15.0	20.0	13.9	17.3	22.7	15.9	20.7	14.6	19.1	3.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	10.8	19.5	13.8	9.9	18.1	12.2	9.0	16.5	49.0	23.1	44.9	20.8	41.2	19.1	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.0	8.5	7.3	-35.3	28.0	4.2	2.6	-38.4	29.9	-40.9	31.5	-43.3	32.9	-46.4	34.9
			-35.5	19.3	4.2	1.7	-38.5	20.5	-41.0	21.5	-43.3	22.5	-46.4	23.7
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-12.0	-12.0	-7.0	-1.2	4.5	9.6	13.4	11.3	7.1	1.0	-5.7	-9.0
	DBStd	10.39	8.05	8.04	6.10	4.64	4.22	4.23	4.06	3.56	3.42	4.39	6.38	7.23
	HDD10.0	3877	683	616	528	336	179	59	13	26	99	278	472	588
	HDD18.3	6681	942	849	787	586	429	265	162	221	337	536	722	847
	CDD10.0	249	0	0	0	0	8	46	117	65	13	0	0	0
	CDD18.3	12	0	0	0	0	0	2	8	2	0	0	0	0
	CDH23.3	132	0	0	0	0	3	19	89	20	0	0	0	0
	CDH26.7	22	0	0	0	0	1	3	17	2	0	0	0	0
(14) Wind	WSAvg	3.2	3.0	3.2	3.4	3.4	3.5	3.5	3.1	2.8	3.1	3.1	3.0	3.1
(15) Precipitation	PrecAvg	486	30	22	23	23	33	54	63	62	52	48	38	36
	PrecMax	659	56	52	53	72	69	142	148	179	120	116	103	113
	PrecMin	277	9	2	5	5	10	3	17	21	19	11	9	3
	PrecStd	81	12	11	12	11	15	29	30	27	23	22	18	18
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	3.7	6.1	12.2	21.2	26.0	28.9	25.9	19.0	11.9	6.1	4.1
		MCWB	1.6	1.9	2.2	6.4	11.9	15.9	18.2	17.3	13.5	9.1	4.9	2.3
	2%	DB	1.0	1.8	3.9	8.7	17.1	22.1	25.9	22.7	16.6	9.9	3.9	2.7
		MCWB	0.0	0.3	1.4	4.4	9.8	13.9	16.7	15.7	12.1	7.8	2.9	1.4
	5%	DB	-0.6	0.2	2.1	6.8	14.6	19.9	23.2	19.9	14.8	8.2	2.6	1.2
		MCWB	-1.5	-0.9	0.0	3.3	8.7	12.7	15.8	14.5	11.5	6.5	1.7	0.2
	10%	DB	-2.0	-1.2	0.8	5.1	11.9	17.2	21.0	17.7	12.9	6.9	1.7	0.0
		MCWB	-2.8	-2.2	-0.8	2.2	7.2	11.5	15.0	13.6	10.3	5.6	0.9	-0.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	2.0	2.7	6.9	12.9	17.3	19.6	18.4	14.1	9.6	4.9	2.5
		MCDB	2.5	3.2	5.3	11.7	20.6	23.7	25.4	24.0	17.6	11.1	5.6	3.7
	2%	WB	0.0	0.4	1.4	4.7	10.6	15.0	18.0	16.5	12.7	8.0	2.9	1.4
		MCDB	0.7	1.5	3.4	7.7	16.4	20.8	23.6	21.0	15.7	9.5	3.6	2.4
	5%	WB	-1.5	-0.9	0.2	3.6	8.8	13.4	16.7	15.2	11.5	6.6	1.6	0.3
		MCDB	-0.7	0.3	1.7	6.2	13.8	18.7	21.8	19.1	14.2	7.8	2.3	1.2
	10%	WB	-2.9	-2.3	-0.8	2.5	7.2	11.9	15.4	13.9	10.4	5.5	0.7	-1.0
		MCDB	-2.1	-1.3	0.6	4.6	11.3	17.0	20.2	17.2	12.6	6.7	1.4	-0.2
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	9.5	9.7	8.4	7.5	8.4	8.8	8.2	7.1	5.1	6.9	8.5
		MCDBR	6.9	6.4	7.3	9.6	13.3	12.8	13.7	12.5	10.4	5.5	4.0	5.9
	5% WB	MCWBR	6.6	5.8	5.8	6.3	7.5	6.5	6.8	6.7	6.8	4.4	3.7	5.4
		MCWBR	7.1	6.3	6.5	8.6	12.4	11.8	12.5	10.8	9.2	5.0	3.8	5.8
(40) Clear-Sky Solar Irradiance	taub	taub	0.206	0.254	0.301	0.332	0.354	0.350	0.368	0.349	0.318	0.277	0.361	N/A
		taud	1.911	2.046	2.098	2.154	2.339	2.446	2.420	2.449	2.497	2.226	2.673	N/A
	Ebn at Noon	Ebn at Noon	38	491	700	797	831	848	815	789	720	471	77	0
		Edh at Noon	6	45	83	107	101	94	94	81	59	38	9	0
(44) All-Sky Solar Radiation	RadAvg	0.02	0.37	1.64	3.29	4.29	4.69	4.37	2.98	1.64	0.47	0.06	0.06	0.00
	RadStd	0.00	0.04	0.09	0.24	0.49	0.38	0.52	0.32	0.17	0.06	0.06	0.01	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page